

A Software-Based Comparative Study of Error Sensitivity in Classical and Quantum Optimization Algorithms

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1 Introduction

Everyone is familiar with the idea of a classical computer, but not everyone is familiar with the idea of a quantum computer. The Classical computing processes information using bits that take definite values of 0 or 1, whereas quantum computing uses qubits that can exist in superpositions of states and become entangled, enabling fundamentally different computational behavior [1]. Quantum computing has the potential to improve the performance of certain computational tasks, particularly search and optimization problems. Many quantum algorithms promise theoretical speedups over classical methods; however, these results often rely on ideal assumptions that ignore noise, limited circuit depth, and other practical constraints of current quantum technology. As a result, there is a gap between theoretical expectations and the real-world behavior of quantum algorithms. This work aims to bridge that gap through a software-based comparison of classical and quantum optimization algorithms, focusing on their sensitivity to noise and errors using classical simulation.

This research evaluates quantum optimization algorithms under realistic execution conditions and compares them with classical heuristics. Rather than focusing on theoretical speedups, it examines practical performance, robustness, and solution quality in noisy environments. Classical methods—genetic algorithms and greedy search—are compared with quantum approaches including Grover-based search and QAOA. A modular benchmarking framework with configurable noise models enables fair, reproducible evaluation using metrics such as solution quality, stability, circuit depth, and noise sensitivity. Results show quantum performance often degrades under realistic constraints, while classical heuristics remain more robust, highlighting the importance of hybrid strategies and practical evaluation methods.

2 Related Works

3 Methodology

Dijkstra’s algorithm was chosen for single-source shortest path computation on graphs with non-negative edge weights, following the implementation presented in CLRS [?].

4 Results

An exemplary figure is shown in Figure 1. Any time a figure is used, it must have a caption and be directly referred to and discussed in the text.

5 Conclusion

References

- [1] WONG, T. G. *Introduction to Classical and Quantum Computing*. Self-published, 2019.

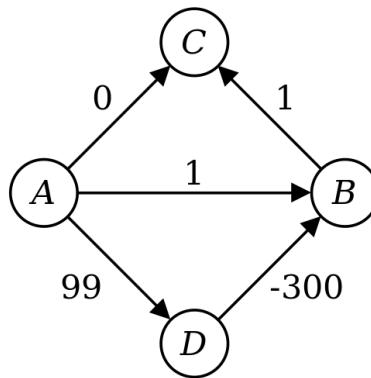


Figure 1: Dijkstra’s algorithm does not work on graphs with negative edge weights, as evidenced here.